

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Monday 26th April 2010

09:30 to 11:30

Year 3 Courses

Convener: K. Kalorloti
External Examiners: K. Eder, A. Frisch

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

1. (a) Distinguish between network and application architectures. [2 marks]
- (b) After a new domain name **broadband4all.net** is created:
 - i. What are the steps involved in registering it with a domain name registrar for the .net domain (e.g., BB Online UK Limited) and inserting a record for **www.broadband4all.net** into the Domain Name System (DNS). [2 marks]
 - ii. Assuming the steps in part (b)(i) are done, explain the DNS name resolution process when a user on the Internet tries to visit the website **www.broadband4all.net**. [2 marks]
- (c) Give five reasons why an application developer might choose UDP over TCP as the underlying transport protocol. [5 marks]
- (d) Distinguish between flow control and congestion control. [2 marks]
- (e) Describe three negative consequences of network congestion. [3 marks]
- (f) Discuss two broad approaches towards congestion control. [2 marks]
- (g) Assume that TCP implements an extension that allows window sizes much larger than 64 KB. Suppose that you are using this extended TCP over a 1-Gbps link with a round-trip propagation delay of 100 ms to transfer a 10-MB file, and the TCP receive window is 1 MB. If TCP sends 1-KB packets (assuming no congestion and no lost packets):
 - i. How many round-trip times (RTTs) does it take until slow start opens the send window to 1 MB? [2 marks]
 - ii. How many RTTs does it take to send the file? [2 marks]
 - iii. If the time to send the file is given by the number of required RTTs multiplied by the RTT value, what is the effective throughput for the transfer? What percentage of the link bandwidth is utilized? [3 marks]

2. (a) List five key differences between IPv4 and IPv6. [5 marks]
- (b) Describe two mechanisms for transition from IPv4 to IPv6. [4 marks]
- (c) BGP and RIP are Internet routing protocols that rely on TCP or UDP for their message exchanges. Since they are conceptually part of the network layer, why do they not just use raw IP packets? [3 marks]
- (d) What is CIDR (Classless InterDomain Routing)? What is the impact of using CIDR on forwarding? [4 marks]
- (e) Suppose a router has the following CIDR entries in its routing table:
- | Net/Prefix | Next Hop |
|-------------------|-----------------|
| 128.96.39.0/24 | Interface 0 |
| 128.96.39.128/25 | Interface 1 |
| 128.96.40.0/25 | Router 2 |
| 192.4.153.0/26 | Router 3 |
| default | Router 4 |
- For each of the following IP addresses, what does the router do if a packet with that address arrives? (i) 128.96.39.10; (ii) 128.96.40.12; (iii) 128.96.39.151; (iv) 192.4.153.17; (v) 192.4.153.90. [5 marks]
- (f) Suppose a large organization with 1000 routers employs a simple 2-level routing hierarchy with the routers equally distributed between 20 areas (as in OSPF) to keep the routing table size small. Assume that, unlike OSPF, every router needs to keep an entry in its routing table indicating the route to reach areas other than the one it belongs to. At what network size in terms of the total number of routers will it no longer be possible to keep the routing table size at or below 100 entries with the above mentioned 2-level hierarchy? [4 marks]

3. (a) Discuss five key differences between circuit and packet switching. [5 marks]
- (b) Suppose that x bits of user data are to be transmitted over a k -hop path in a packet-switched network as a series of packets, each containing p data bits and h header bits, with $x \gg p + h$. The bit rate of the lines is b bps and the propagation delay is negligible. What value of p minimizes the total delay? [4 marks]
- (c) Datagram networks route each packet as a separate unit, independent of all others. This is not the case with virtual-circuit networks, where each data packet follows a predetermined route. Does this observation mean that virtual-circuit networks cannot route isolated packets from an arbitrary source to an arbitrary destination? Explain. [2 marks]
- (d) As each network interface at a router or host is assigned a unique IP address, why is a separate MAC address (also called physical address) needed? [2 marks]
- (e) When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss the relative merits of these two approaches. [4 marks]
- (f) Suppose the data portion (including the headers) of a packet contains a sequence of 8 bytes, each corresponding to 8-bit unsigned binary representation of the following integers: 243, 81, 128, 64, 32, 16, 27, 9. Compute the Internet checksum for this data. [4 marks]
- (g) Consider the 4-bit generator, $G = 1001$, and suppose that data portion of the bit stream to be transmitted prior to Cyclic Redundancy Check (CRC) calculation is 11001001.
- Show the actual bit stream transmitted. [2 marks]
 - Suppose the leftmost bit in the transmitted bit stream is inverted due to noise on the transmission link. Show that this error is detected at the receiver's end. [2 marks]